Significance:

Sometimes inversion is responsible for evolution of the organism.

NOTE: Likewise the teacher can give different types of chromosomal aberrations with various gene sequence to students for practise. The external examiner can also use the same technique by giving different gene sequence.

Exercise 14: Genetic / linkage maps

Aim:

To understand the frequency of recombination between the gene pairs on the same chromosome.

Principle:

To analyse the relative distance between the various genes and map their position in the chromosome, which is called genetic or linkage maps.

Requirements:

Different kinds of linkage / genetic maps can be constructed by giving the students the relative distance between the linked genes of a chromosome. A diagrammatic representation can be drawn showing the location and arrangement of genes and their relative distance between them.

Solve the Problem

Problem: There are three linked genes A, B and C in a chromosome. Percentage of crossing over (recombination frequency) between A and B is 20, B and C is 28 and A and C is 8. What is the sequence of genes on the linkage map?

Given: Percentage of crossing over between the 3 linked genes A – B = 20%, B – C = 28% and A – C = 8%.

Solution

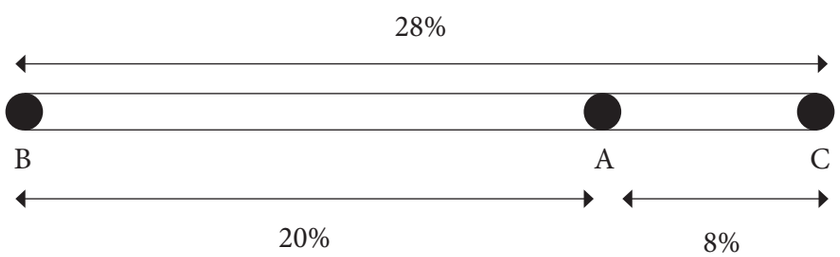

Figure 14: Linkage Map

Reasons:

1. The frequency of crossing over is directly proportional to the relative distance of the genes on the chromosomes.
2. More crossing over = More distance between two genes and
   Less crossing over = Less distance between the two genes.

In the above problem, the sequence of the genes on the linkage map is B, A, C

NOTE: Teachers can give different crossing over percentage between its linked genes in a chromosome and make the students construct the linkage maps. The external examiner can also do the same for the Board Practical Examinations.

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